



Aspect Calc user guide

Aspect Calc is an **aspect ratio, pixel pitch and display geometry calculator for people who put pictures on walls**. Browser only — no account, no backend, nothing you type leaves the tab.

The whole model is one relation:

$$\text{resolution (px)} \times \text{pixel pitch (mm)} = \text{physical size (mm)}$$

Give it any two and it calculates the third. Diagonal and aspect ratio are two more ways of writing the physical size, so you can also hand it a 55" diagonal and "16:9" and get the width and height back.

Before you rely on this: the arithmetic is pinned by 100 tests, including the ratio-naming rules — which are the only part with judgement in them.

Nothing here has been checked by measuring a real screen, because there is nothing to measure: it is arithmetic over the numbers you type, and it is only as right as they are.

This codebase was created with AI assistance, directed and reviewed by a human author.

What it does

- **Aspect ratio from anything** — a resolution in pixels, a physical size in mm/cm/m, inches or feet-and-inches, or a diagonal and a ratio.
- **Pixel pitch in any direction** — pitch from resolution and size, size from resolution and pitch, or resolution from size and pitch. **Non-square pixels are reported, not averaged away.**
- **Both decimal forms**, `1.778 : 1` and `0.5625`.
- **Diagonal and area**, always, in both measuring systems at once.
- **PowerPoint slide sizes**, both directions.
- **The display drawn to scale**, with colour bars and every dimension on it.

It names the ratio properly

3440×1440 reduces exactly to **43:18**, and nobody has ever ordered a 43:18 monitor — it is called **21:9**. 1366×768 reduces to 683:384, which is not a ratio so much as an accusation; the useful answer is **16:9, 0.05% off**.

Both the trade name and the arithmetic are always shown, and a match that is not exact says so and says by how much. That last part matters: a ratio quoted as exact when it is 0.05% off is how a picture ends up one pixel short across a wall.

Three things from the trade

Pitch is centre-to-centre and every pixel owns a full cell, so physical width is $\text{horizontal pixels} \times \text{pitch}$, **not** $(\text{pixels} - 1) \times \text{pitch}$.

A 168×168 px cabinet at 2.9 mm is **487.2 mm** square, not 484.3. Getting this wrong loses one pitch across the whole wall — **invisible in a spreadsheet and very visible when the last cabinet does not fit the frame**.

The two 21:9s are not the same ratio. 2560×1080 is 64:27 (2.370:1); 3440×1440 is 43:18 (2.389:1). They are **0.8% apart** and both are sold as "21:9". The calculator names them both 21:9 and then tells you which one you have.

The colour bars are a picture, not a signal. The pattern is stretched to whatever aspect you are looking at — which is what a real generator does — and the RGB values are the standard 75% bar values but **are not colour-managed. Do not grade against it**.

PowerPoint

Both directions: the slide size for a target resolution, or the resolution a given slide size exports to.

It handles the format's own limits — the **56-inch slide cap**, the 1-inch floor and the 100 MP export ceiling — and gives you points and EMU alongside inches and centimetres, because those are the units the file format actually stores.

If a number looks wrong

A ratio is named 16:9 but the numbers do not reduce to it. That is deliberate — the trade name and the exact reduction are both shown, with the error between them.

A wall comes out one pitch too small. You are using $(\text{pixels} - 1) \times \text{pitch}$. See above.

The pitch differs horizontally and vertically. The pixels are not square, and the tool reports that rather than averaging it away. That is a real property of some LED products.

PowerPoint refuses the slide size. It is past the 56-inch cap. The calculator says so rather than producing a number the format cannot hold.

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